

CHAPTER FIVE : MORPHOLOGY OF FLOWERING PLANTS

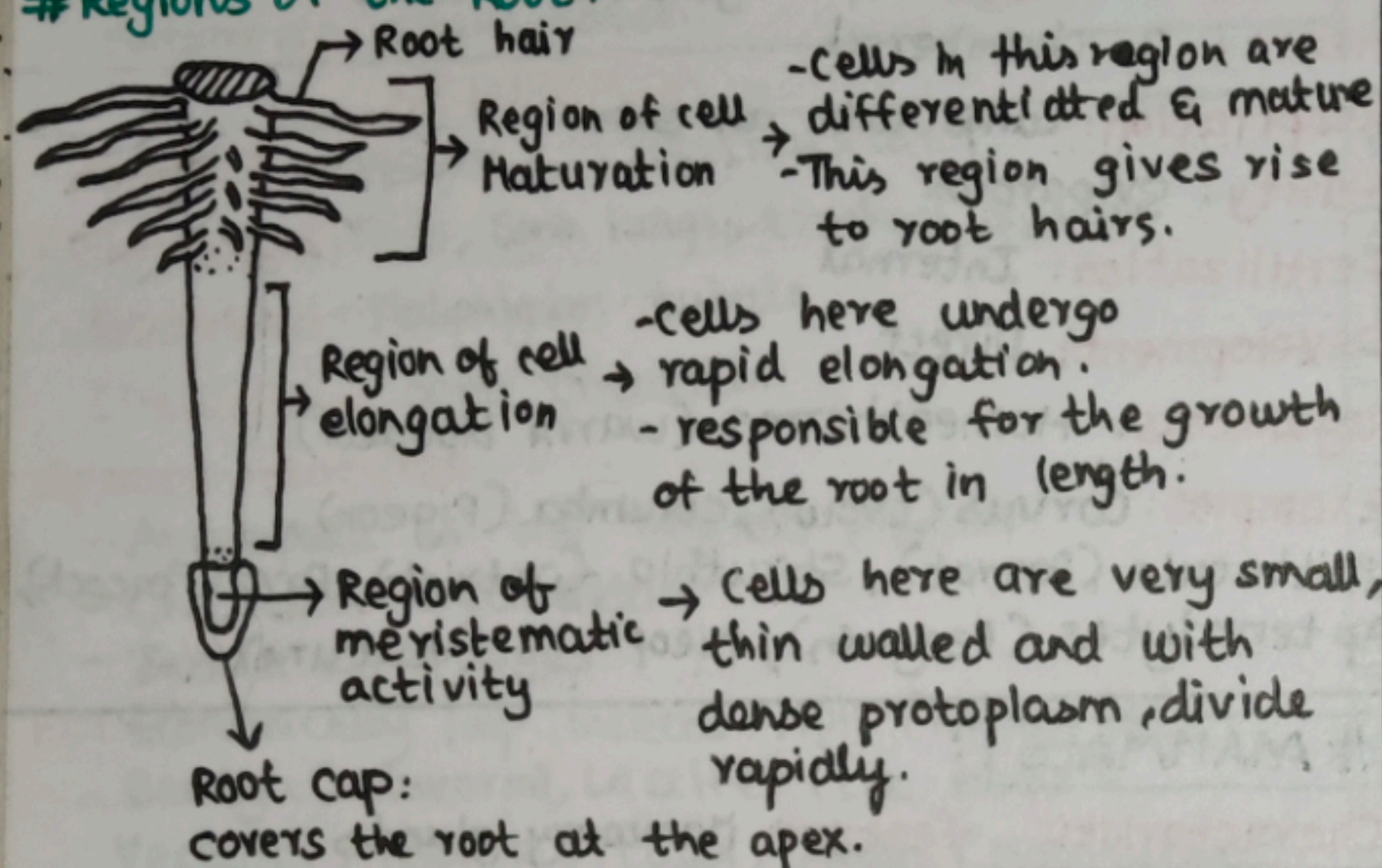
* Katherine Esau: received awards for work on plant anatomy.
 * Morphology refers to ext. structure.

THE ROOT

Types of Roots:

Tap roots	Fibrous roots	Adventitious roots.
Radicle gives rise to 1 root which grows inside soil. Branches of 1 root forms 2, 3, etc roots	1 root is short lived. It is replaced by roots which originate from the base of the stem.	Roots arise from parts of the plant other than the radicle.
Dicot root	Monocot root	Girans, Monstera, Banyan tree

Regions of the Root:



Modifications of Root:

- Tap root (storage) - carrot, turnip.
- Adventitious root (storage) - Sweet potato, Asparagus.
- Support
 - Prop root - Banyan tree
 - stilt root - Maize, sugar cane.
- Respiratⁿ (Pneumatophores) - Rhizophora (Marshy area and Halophytes).

Functions of Roots:

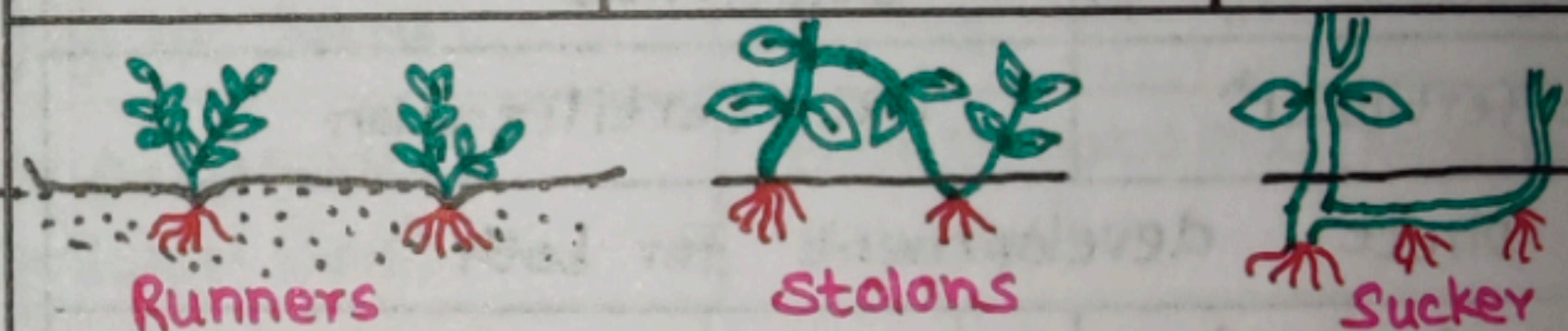
- Food storage
- Providing a proper anchorage to the plant.
- Absorption of water and minerals.
- Synthesis of plant growth regulators (Auxin & cytokinins)

THE STEM

- * Develops from the plumule.
- * The stem bears nodes and internodes
 - Nodes - leaves are born at nodes
 - Internodes - portions b/w two nodes
- * Stem bears buds.
- * Apical buds: Involve in 1st growth of plant
- * Axillary (lateral) buds: develops into branches or flowers, leaf petiole is attached here.

Modifications of stem:

Modified stem	Function	Examples
Tendrils (axillary buds)	Climber	Cucumber, pumpkin, watermelon, grapevines
Stem swollen	Food storage	Potato, ginger, turmeric, zaminkand, colocasia.
Thorns (axillary buds)	Protection	Citrus, Bougainvillea
Flat stem (Phylloclade)	Photosynthesis	Opuntia
Fleshy/cylindrical stem	Photosynthesis	Euphorbia
Underground stem (Runner)	Spread to new niche	grass and strawberry
Stem arch downward (Stolon)	Spread of plant	Mint and Jasmine
Lateral branch nodes forms leaf and roots (offset) in water	Spread of plant	Pistia and Eichhornia.
Lateral underground branches (suckers)	spread of plant	Banana, pineapple, chrysanthemum



THE LEAF

- Develops at the node and bears a bud in its axil. The axillary bud later develops into a branch.
- Leaves originate from shoot apical meristem.
- Arrangement of leaf: Acropetal order.

Parts of leaf:

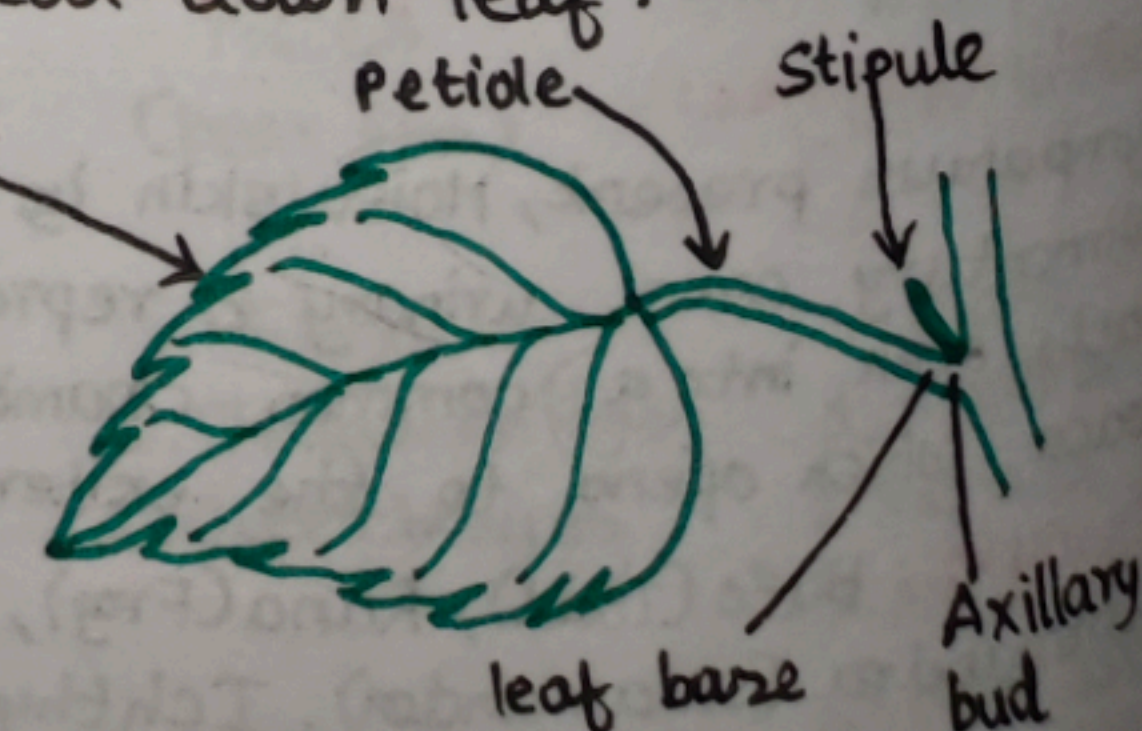
→ Leaf base:

- Leaf is attached to the stem by the leaf base.
- Leaf base bears small leaf like structures called stipules.
- Monocots: Sheath covers leaf base.
- Leguminous plants: Swollen leaf base (pulvinus)

→ Petiole: Help hold the blade to light - flutter with wind - cool down leaf.

→ Lamina or the leaf blade:

- Have middle prominent vein (mid rib) and veinlets



Types of leaf:

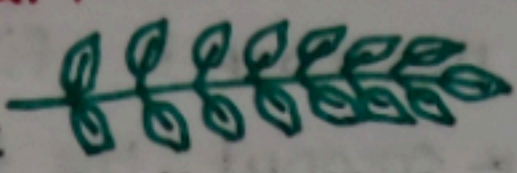
→ **Simple leaf**: Lamina is entire or when incised, the incisions do not touch mid-rib.

→ **Compound leaf**: Incisions of the lamina reach up to the midrib breaking it into a no. of leaflets.

Types of compound leaf:

i) Pinnately compound:

→ leaflets are present on a common axis, midrib called as Rachis.



eg: Neem.

ii) Palmately compound:

→ The leaflets are attached to petiole.

→ Eg: silk cotton



Phyllotaxy: Pattern of arrangement of leaves on the stem or branch.

Alternate	Opposite	Whorled
→ Single leaf arises at each node in alternate manner.	→ Pair of leaves arise at each node & lie opp. to each other.	→ More the 2 leaves arise at a node and form a whorl.
→ Sunflower, Mustard, China rose	→ Calotropis, guava	→ Alstonia.

Modification of leaves:

→ **Leaf Tendril**: Peas

→ **Leaf Spines**: Cacti

→ **Fleshy Leaves**: Onion, Garlic

→ **Photosynthetic Petiole**: Australian acacia

→ **Insectivorous plant**: Pitcher plant, Venus fly trap

THE INFLORESCENCE

→ Arrangement of flower on floral axis.

Racemose	Cymose
Floral axis have indefinite growth	Floral axis terminates into flower.
Old flower at base- young flower at top	Old flower at top young at base
Acropetal - Fabaceae, Brassicaceae.	Basipetal

THE FLOWER

→ The flower is the reproductive unit in the angiosperms.

→ 4 parts: calyx, corolla (Accessory organs), androecium, gynoecium (Reproductive organs)

Bisexual flower: Both stamen and pistil present.

Unisexual flower:

→ **Staminate flower**: Stamen only present

→ **Pistillate flower**: Pistil only present

Symmetry	Definition	Example
Actinomorphic (Radial symm.)	Flower can be divided into 2 equal radial halves by any radial plane.	Mustard, Datura, Chilli (MCD).
Zygomorphic (Bilateral symm.)	Flower can be divided into 2 equal radial halves by single plane	Pea, Gulmohur, Bean, Cassia (Pea GPS)
Asymmetric (irregular)	No plane divides flower into 2 similar halves	Canna

* **Hypogynous** - Superior ovary - Mustard, China, Rose, Brinjal

* **Perigynous** - Half inferior ovary - Plum, Peach, Rose

* **Epigynous** - Inferior ovary - Guava, cucumber, Ray florets of sunflower

* A flower may be trimerous, tetramerous or pentamerous when the floral appendages are in multiple of 3, 4 or 5, respectively.

* Flowers with **bracts** - reduced leaf found at the base of the pedicel are called **bracteate** and those without bracts, **ebracteate**.

Parts of a flower:

1. **Calyx**: → outermost whorl of the flower.

→ Members are called sepals. Generally sepals are green.

→ Types: **Gamosepalous** - sepals united
polysepalous - sepals are free

2. **Corolla**:

→ Members are called petals.

→ Petals are usually coloured to attract insects for pollination.

→ Types: **Gramopetalous** - petals united
Polypetalous - petals free

→ **Perianth**: calyx & corolla are not distinct. eg: lily.

⇒ **Aestivation**: Arrangement of sepals or petals wrt each other observed in bud stage.

Aestivation	Margin type	Example	Diagram
Valvate	Margin do not overlap	Calotropis	
Twisted	Twisted	China rose, lady's finger, Cotton	
Imbricate	Margin overlap in irregular manner	Cassia, Gulmohur	
Vexillary or papilionaceous	Largest petal (standard) overlaps the 2 lateral petals (wings) which in turn overlap two smallest anterior petals (keel)	Pea, Bean.	

CHARACTER	BRASSICASSIA	MALVACEAE	LEGUMINOSAE ^{real family}	ASTERACEAE	POACEAE ^{Grass family}
Floral Formula	$Ebr \oplus \overset{\circ}{\underset{\circ}{\text{K}}}_{2+2} C_4 A_{2+4} \overline{G}_{(2)}$ Cruciferae (Mustard family)	$Br \oplus \overset{\circ}{\underset{\circ}{\text{K}}}_{(5)} C_5 A_{(5)} \overline{G}_{(5)}$ Hibiscus family	* Fabaceae: $\% \overset{\circ}{\underset{\circ}{\text{K}}}_{(5)} C_{1+2+(2)} A_{(1)+1} \overline{G}_1$ * Caesalpiniaceae: $Br \% \overset{\circ}{\underset{\circ}{\text{K}}}_5 C_5 A_{3+4+3} \overline{G}_1$ * Mimosaaceae: $Br \overset{\circ}{\underset{\circ}{\text{K}}}_4 C_4 A_4 \overline{G}_{11}$	* Disc floret: $Br \oplus \overset{\circ}{\underset{\circ}{\text{K}}}_{pappus} C_{(5)} A_{(5)} \overline{G}_{(2)}$ * Ray floret: Sunflower family $Br \% \overset{\circ}{\underset{\circ}{\text{K}}}_{pappus} C_{(5)} A_0 \overline{G}_{(2)}$	$\% \overset{\circ}{\underset{\circ}{\text{K}}}_2 P_2 (Lodicules) A_{3+3} \overline{G}_1$
Inflorescence	Racemose	Solitary	Racemose	Special racemose: Head	Spike or panicle
Flower	Complete, Bisexual, actinomorphic, hypogynous	Complete, Bisexual, actinomorphic, hypogynous	Complete, Bisexual, actinomorphic / zygomorphic Hypogynous / perigynous	Epigynous, Flowers are called florets Disc floret Ray floret	→ Flower with branch of spike called spikelets. → Incomplete flower → Bisexual, Zygomorphic, Hypogynous
Calyx	Sepals (4), polypetalous	→ Sepals (5) gamasepalous → outer to sepals a whorl of bract present: Epicalyx	→ Sepals (4 or 5) → gamo / polypetalous → valvate / imbricate	→ Sepals modified as pappus.	Undifferentiated.
Corolla	Petals (4), polypetalous valvate, cruciform.	Petals (5), polypetalous Imbricate.	Petals (4 or 5), polypetalous vexillary or imbricate	Petals (5), gamopetalous Valvate	undifferentiated. Lodicules represent perianth.
Androecium	Stamen (6), Tetrastamens (2+4)	→ Stamens numerous → Monadelphous. → Monotheus & Reniform anther.	Stamen (10) Diadelphous / Polyandrous.	Stamen (5) - show • cohesion (syngenesious) • adhesion (epipetalous)	→ Stamens (3-6) polyandrous Anther - versatile.
Gynoecium	Bicarpellary, syncarpous	Multicarpellary, syncarpous	Monocarpellary, unilocular	Bicarpellary, syncarpous	Tricarpellary, syncarpous
Placentation	Parietal placentation	Axile placentation	Marginal placentation	Basal Cypsella	Basal Caryopsis
Fruit	Silique	Capsule	Legume	Non-endospermous	Endospermous
Seed	Non-endospermous	Endospermous	Non-endospermous	Herbs & shrubs	Annual and Perennial herbs
Habit	Mostly Annual / Perennial Herbs.	Herbs & shrubs	Herbs / shrub / tree	Tap root	Fibrous root
Root	Tap root	Tap root	Tap root	Erect, trichomes present	Culm (solid nodes & hollow internodes)
Stem	Erect with reduced internodes.	Erect, mucilage secreting hairs are present.	Erect / Prostrate (climbers)	simple, exstipulate	simple, exstipulate with sheathing base, parallel venation.
Leaf	Simple, exstipulate, petiolate, alternate, reticulate venation	Simple, stipulate, petiolate, alternate, reticulate venation	compound, commonly pinnate pulvinate		